10.7.6 Treatment of PEM

Children with severe PEM are often seriously ill when they first present for treatment. They should be admitted to a hospital for the treatment of life- threatening problems. Specific deficiencies should be corrected and metabolic abnormalities reversed. When the child's condition is stable and the appetite has returned,which is usually after 2- 7 days ,the treatment can be continued outside the hospital .

A. Hospital based management

1) **Dehydration:** Diarrhoea leading to dehydration is a serious and often fatal event in children with severe malnutrition. Skin elasticity is poor in children with marasmus and their eyes are normally sunken. Unlike Kwashiorkor, the altered skin elasticity is masked by oedema. Patients with mild to moderate dehydration can be treated by oral or nasogastric administration of fluids.

2) **Infection:** Infection is often the immediate cause of death in PEM. It is difficult to detect infections clinically as fever and rapid pulse rate may not be present in severely malnourished patients. Since infection is common, antibiotics should be given routinely to all malnourished patients. Children with complications should be treated with broad spectrum antibiotics like amoxicillin and ampicillin. Intestinal infections like ascariasis must be treated with appropriate de-worming agents.

3) **Hypoglycemia:** A child may become drowsy or develop convulsions due to hypoglycaemia. In mild cases, oral administration of 50 ml of 10% glucose may be sufficient. If a child develops convulsions or becomes unconscious, 10% glucose should be given intravenously (5ml/kg) followed by 50ml of 10% glucose by nasogastric tube.

4) **Hypothermia:** Marasmic children are prone to have low body temperature. If the room is cold, the child should be properly covered with a blanket. The state of shock should be treated with intravenous injection of glucose –saline or blood transfusion.

5) **Anaemia:** Severe anaemia is dangerous, as it can result in heart failure. If the haemoglobin falls below 5g/dl, blood transfusion should be given.

B. Dietary management

Although treatment of complications can reduce mortality, proper dietary management is important for complete recovery. The child should be given a diet providing sufficient quantities of calories and protein, in gradually increasing amounts, without provoking vomiting or diarrhoea. It is best to begin with liquid formula, as it is easy to feed and measure the intake. Initially the child may refuse the feeds due to lack of appetite. As the appetite improves and child starts taking food by mouth, solid supplements can be introduced. The diet should be given frequently and in small amounts.

High energy intakes (150Kcal/kg) and high protein intakes (3-4g/kg) are required for rapid recovery. Most hospitals use milk-based formulas for feeding

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